



Mid Term Project Presentation

on

AGRI-SHIELD AI: A Deep Learning and IoT-based Framework for Automated Crop Insurance Claims.

By

Aayush Vijay(2023kuec2068)

Naman jain(2023kuec20xx)

Hansraj(2023kuec20xx)

Kushagra Bhardwaj(2023kuec20xx)

Under the supervision of

Dr. Ashok Kumar Kherodiya

Department of ECE

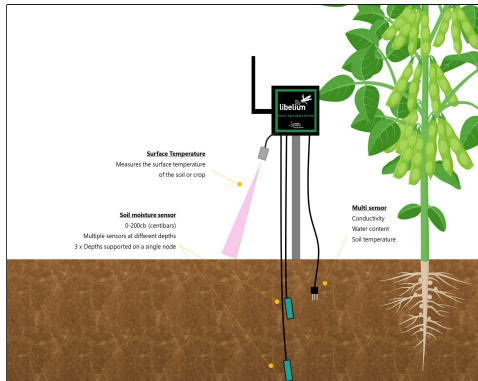


Problem Statement & Objectives

- **90+ Days Average Settlement Delay (NITI Aayog Data):**
 1. Relies heavily on traditional "Crop Cutting Experiments" (CCEs) which are labor-intensive and cannot be scaled during widespread natural disasters.
 2. Payouts often arrive too late for farmers to replant for the next season, leading to debt cycles.
- **₹20,000 Crores in Annual Estimated Fraud:**
 1. Manual audits are highly susceptible to bribery and subjective human assessment.
 2. Bad actors frequently submit fabricated, recycled, or geographically spoofed photographic evidence to claim false losses.

The Triangulated Truth Model (Core Concept)

Ground Truth (Hardware IoT)



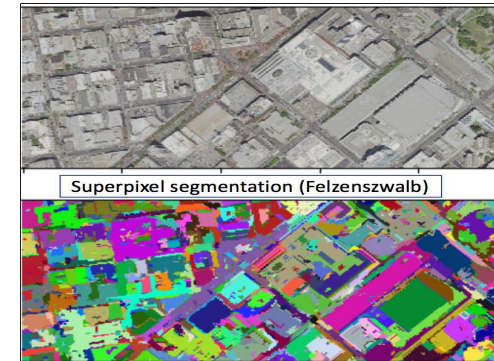
Edge nodes buried at the root-zone level capture continuous, real-time telemetry including volumetric soil moisture, temperature, and humidity.

Visual Truth (Machine Learning)



A customized ResNet-50 Convolutional Neural Network (CNN) validates visual crop damage via an app, requiring geotagged photos and gyroscope-enforced 360° video.

Macro Truth (Satellite Data)



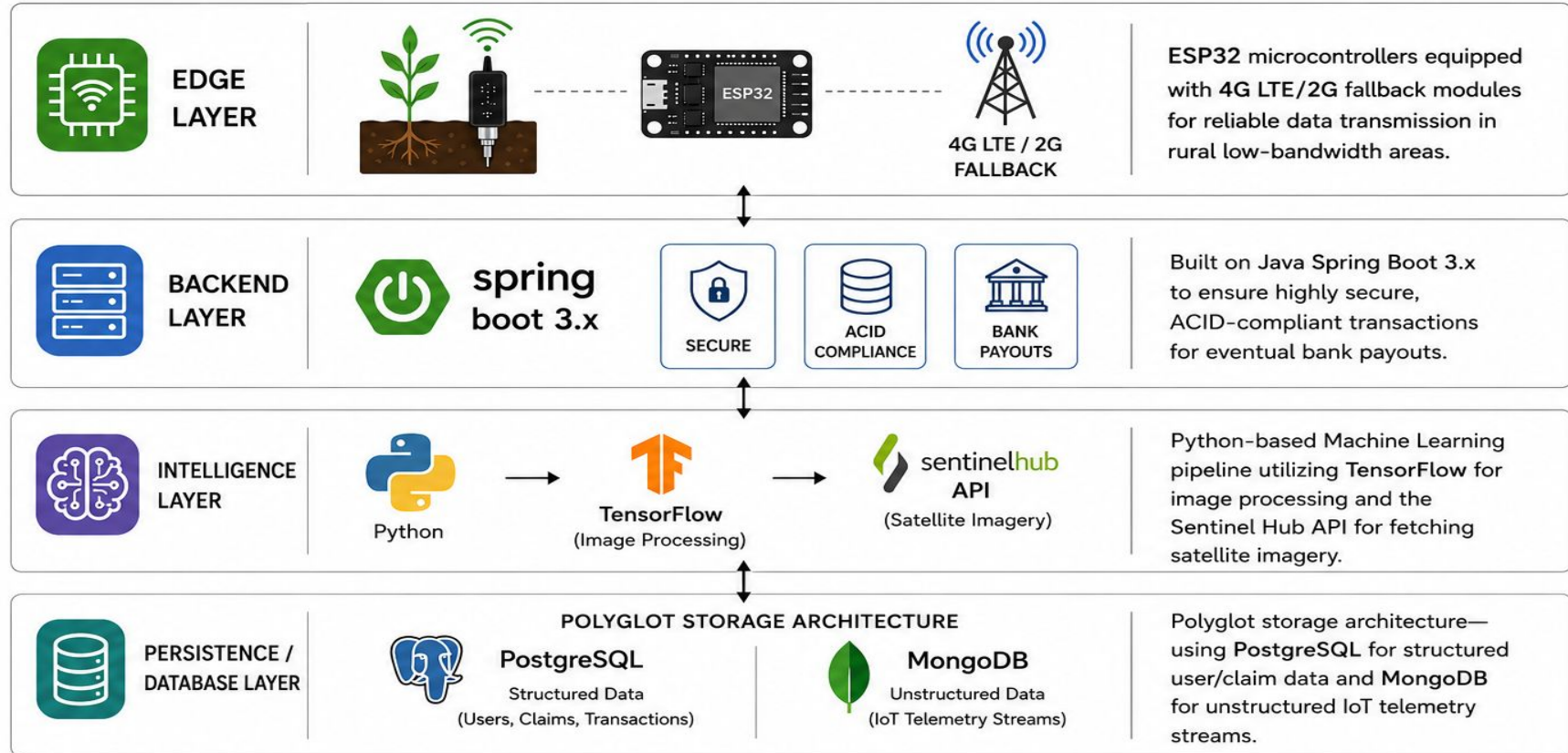
Sentinel-2 multispectral imagery computes the Normalized Difference Vegetation Index (NDVI) to verify regional, large-scale vegetation stress trends, completely removing localized human bias.

1. <https://manxtechgroup.com/soil-monitoring-with-iot-smart-agriculture/>
2. <https://www.rtb.cgiar.org/wp-content/uploads/2017/09/Pic-4-1-1024x785-1.jpg>
3. <https://forums.fast.ai/t/object-detection-in-satellite-imagery-which-approach/12055>

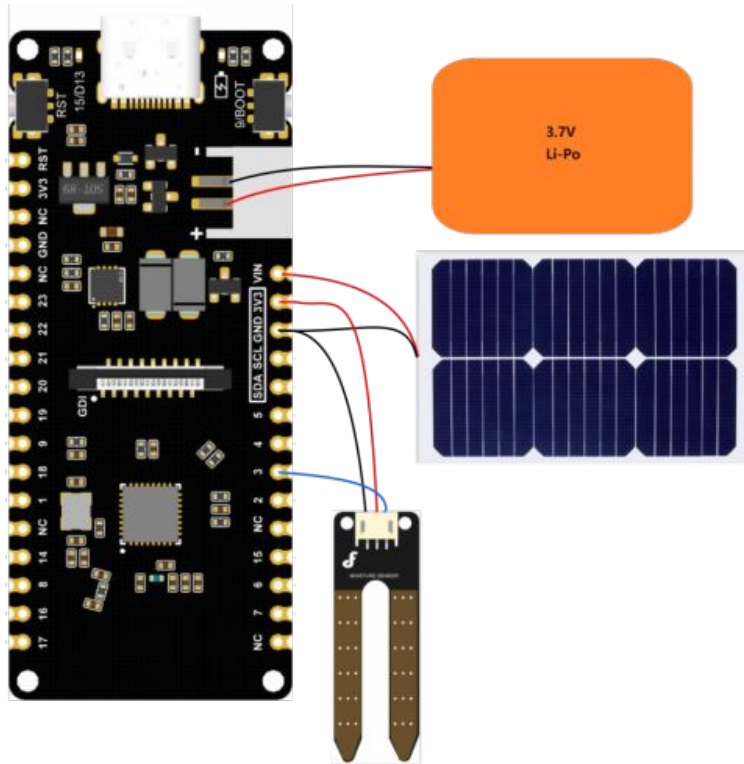
Full-Stack System Architecture



Micro-serviceable Modular Monolith Design



IoT Edge Node: Hardware Engineering



Engineered for Harsh Environments:

The ESP32 dual-core processor efficiently handles both sensor polling and cellular network handshakes. We utilize capacitive soil sensors to entirely prevent the electrolytic corrosion that destroys traditional resistive sensors in salty, heavily fertilized soils.

Strict Power Budget Management:

- Operates on an energy-neutral basis using a 5V/2W solar panel paired with dual 18650 Lithium-ion batteries.
- Uses extreme Deep Sleep cycles (consuming roughly 22.5 mAs) to wake, transmit data, and return to sleep, ensuring months of maintenance-free operation.

Multi-Layered Anti-Fraud Security Framework



1 Mock Location & Geofencing Defense



The app bridges native Android/Java APIs to detect and block GPS spoofing applications. It uses PostGIS `ST_Within()` functions to strictly validate that the user's GPS coordinates fall within their officially registered farm boundaries (Khasra polygons).

2 Gyroscope-Enforced 360° Pan



To prevent users from simply recording a video of a static computer screen showing damaged crops, the app forces a mandatory 5-second video recording that requires the smartphone's gyroscope to register a physical 270° angular sweep of the environment.

3 XGBoost Anomaly Detection



The system cross-references the physical soil moisture sensor data against historical and forecasted precipitation data (via OpenWeather API). If a sensor reads "flooded" but no rain has fallen, it flags the sensor for physical tampering (e.g., a farmer pouring a bucket of water on the sensor).

1. <https://www.filecloud.com/blog/geo-fencing-in-data-governance-and-its-possible-uses/>
2. <https://community.iqoo.com/in/thread/55641>
3. <https://www.labellerr.com/blog/content/images/2024/01/Anomaly-Detection-CV-Main.webp>

ML Performance: Visual Damage Classification



Model Accuracy Comparison

(Validation Set)

ResNet-50 (Custom Fine-tuned) (Chosen Model)	96.2%
InceptionV3	91.4%
MobileNetV2 (Faster, but less accurate on edge cases)	89.1%
VGG-16	87.2%

Training Insights

Generic agricultural datasets were insufficient. We achieved the **96.2% accuracy** by fine-tuning the ResNet-50 model specifically with **8,240 custom photographs** of Indian fields, allowing the AI to learn domain-specific damage morphology (e.g., locust damage, localized flooding, drought cracking).



The complete flow after this

[https://www.ultralytics.com/blog/wh](https://www.ultralytics.com/blog/what-is-resnet-50-and-why-it-matters-in-computer-vision)
[https://www.youtube.com/watch?v=](https://www.youtube.com/watch?v=mgMphYiN5lk)

Claim Initiation & Local Verification (The Frontline Flow)



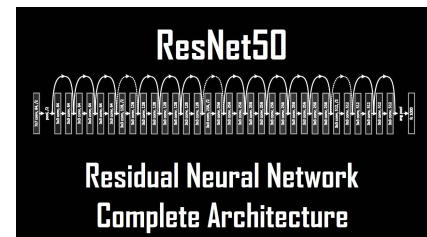
1 **Onboarding & Land Validation:** Farmers log in via Aadhaar/Mobile OTP. Their *Khasra* (land parcel) is digitally verified by the local official (Patwari App). *Note: If the Patwari delays, an automated grievance is sent to HQ.*



2 **Disaster Reporting & Evidence Capture:** Immediately after crop damage, the farmer opens the app to initiate a claim. To prevent fraud, the app enforces a mandatory 360° gyroscope-tracked video and captures geo-tagged, timestamped photos on the exact verified coordinates.



3 **Instant Edge AI Assessment:** The uploaded visual evidence is immediately processed by the embedded Machine Learning model (ResNet-50). The AI identifies the specific damage morphology and generates a preliminary "Damage Severity Score."



Triangulated Audit & Automated Payout (The Intelligence Flow)



Step 2: Triangulated Verification & Admin Resolution

- **Ground & Weather Cross-Check:** Before trusting the AI score, the system automatically correlates the claim with data from buried IoT soil sensors and regional Weather APIs (e.g., verifying that a flood claim matches physical soil moisture spikes and rainfall history).
- **Satellite Proof (Macro Truth):** The system fetches Sentinel-2 satellite imagery to calculate the NDVI score. A mathematical decline in vegetative health compared to the farm's historical baseline acts as un-fakable proof of crop stress.
- **Admin Dashboard & DBT Payout:** All data streams (AI + Sensors + Weather + Satellite) merge into the centralized **Admin Dashboard**. Fully corroborated claims bypass manual red tape and instantly trigger a Direct Benefit Transfer (DBT) payout to the farmer's bank account within 24 hours.

Project Status & Future Scope



Current Implementation

End-to-End System Prototype:

- Frontend UI: Fully functional Farmer Mobile App (vernacular interface) and secure Web Dashboards for Patwaris and HQ Admins.
- Backend & Intelligence: Robust server infrastructure (Java Spring Boot) integrated with the custom ResNet-50 visual damage assessment model.
- Hardware Simulation: ESP32-based IoT node prototypes actively simulating real-time soil telemetry and edge-to-cloud data handshakes.

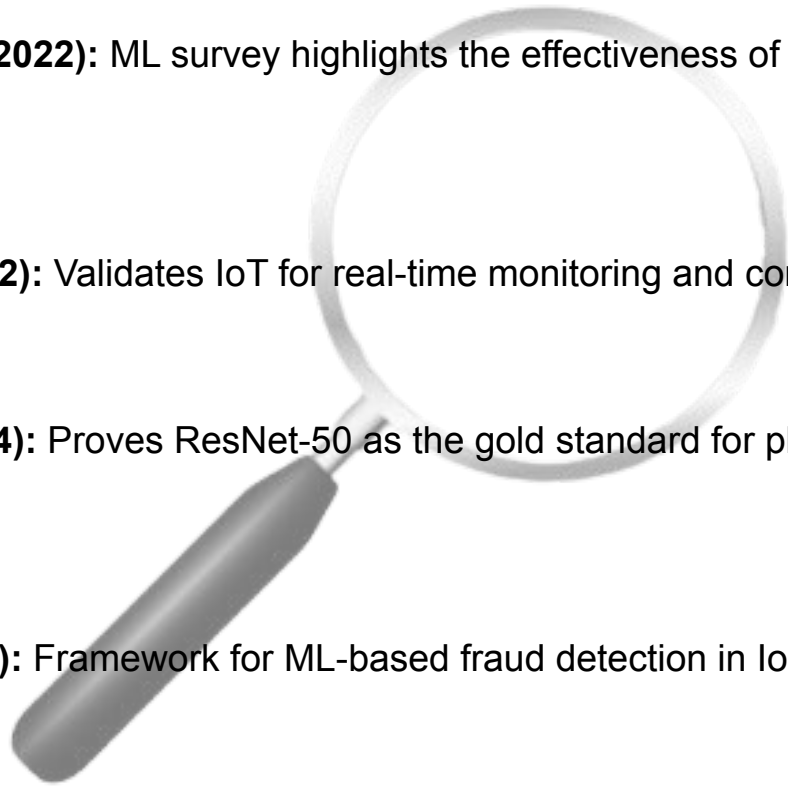
Future Scope

- **Phase 2 - Network Expansion:** LoRaWAN integration to ensure continuous sensor connectivity in ultra-remote agricultural zones that lack reliable 2G/4G cellular coverage.
- **Phase 3 - Immutable Security:** Implementation of a Blockchain Audit Trail (using Hyperledger) to make government claim records and AI decisions mathematically tamper-proof.
- **Phase 4 - National Integration:** Deep system integration with the official PMFBY (Pradhan Mantri Fasal Bima Yojana) national portal and automated Banking APIs for direct DBT payouts at a national scale.

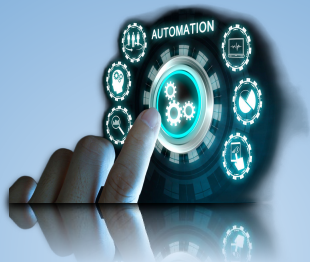
References



LITERATURE REVIEW

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- **Domingues et al. (2022):** ML survey highlights the effectiveness of CNNs in crop disease prediction.
 - **Rehman et al. (2022):** Validates IoT for real-time monitoring and control in smart agriculture.
 - **Rani & Gupta (2024):** Proves ResNet-50 as the gold standard for plant leaf disease detection (98.4% accuracy).
 - **Sharan et al. (2024):** Framework for ML-based fraud detection in IoT-linked insurance claims.

RESEARCH GAP IDENTIFIED



Lack of End-to-End Systems: Most research focuses only on "detecting" disease, not on the "financial payout" workflow.

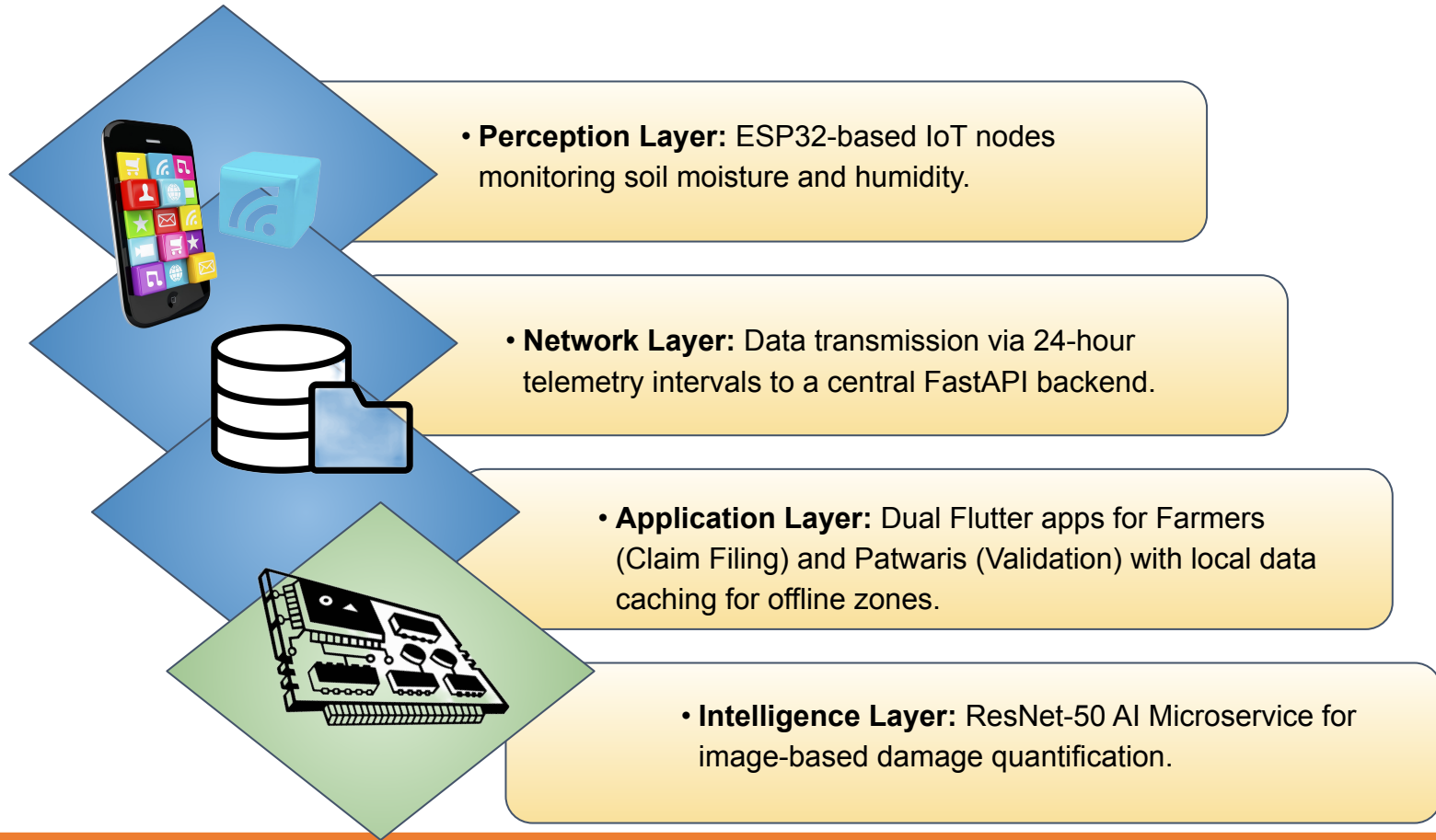


Verification Void: Existing systems often lack ground-truth sensor data to verify the "environmental cause" of reported damage.



Satellite Limitations: Satellite-based claims are often blocked by cloud cover or low spatial resolution; our ground-level AI solves this.

Proposed System Architecture





Fraud Prevention Strategy

- **Geofence Verification:** App captures Lat/Long; photo uploads are blocked if the user is $>10\text{m}$ from the land's center.



- **Time-Series Correlation:** AI damage assessment must correlate with the 24-hour IoT moisture/humidity history to prevent fake photos.



- **Metadata Validation:** Automatic timestamp and device ID mapping for every claim.



Technical Stack

Mobile: Flutter / Dart (Cross-platform Farmer & Patwari apps with SQLite/Hive for offline sync).

Backend: Java / Spring Boot 3.2 (RBAC and business logic).

AI/ML: Python / FastAPI / PyTorch (ResNet-50 Model).

Hardware: ESP32, Capacitive Soil Sensors, DHT11.

Database: PostgreSQL (Structural Land & User data).



Current Status - Frontend Progress

- **Farmer App:** Registration, Aadhar-auth, and Land Selection (Khasra No) screens completed.
- **Patwari App:** Dashboard showing "Pending Claims" and "Field Inspection" queue is ready.
- **Payment Gateway:** Razorpay "Test Mode" successfully integrated for premium payments.



Current Status - Backend & AI

- **API Development:** User authentication and Land Record mapping APIs are functional.
- **Database:** Schema for "50 data points" (Land, User, Claims, Sensors) is implemented.
- **AI Training:** Initial training on the PlantVillage dataset (38 categories) has begun.

